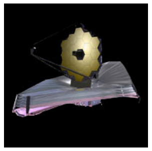

10108 - Probing shock driven particle acceleration using near infrared observations

Cycle: 5, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
NIRcam				
	1	NIRcam H and [Fe II]	NIRCam Imaging	(3) East-West_lobes
NIRSpec IFU				
	2	NIRSpec IFU for east lobe	NIRSpec IFU Spectroscopy	(1) East_Lobe
	3	NIRSpec IFU for west lobe	NIRSpec IFU Spectroscopy	(2) West_Lobe

ABSTRACT

Protostellar jets are highly collimated material launched from young stellar objects during early phases of star formation. When fast-moving protostellar jets interact with the ambient medium, they generate regions of gas, some of which show thermal free–free emission, a few exhibit non-thermal synchrotron radiation, indicating relativistic particle acceleration. The physical shock conditions enabling such acceleration, however, remain poorly understood. We propose JWST/NIRSpec and NIRCam observations of the massive protostellar jet system RAFGL2591 to directly link shock physics with the onset of non-thermal emission. RAFGL2591 uniquely hosts both thermal and non-thermal jet lobes, offering an unparalleled opportunity to compare their shock environments. Using near-IR diagnostics lines, we will constrain temperature, electron density, ionization

fraction, and extinction across both lobes. The NIRCam imaging will reveal asymmetry in forward and reverse shocks, distinguish between C- and J-type shocks and determine where the particle acceleration arises using H and [FeII] filters. Simulations with the Paris–Durham shock code predict that C-type shocks produce strong molecular H emission, while J-type shocks yield enhanced ionic lines—providing testable signatures for our proposed spectroscopy. By comparing modeled and observed line ratios, and excitation diagrams we will determine whether synchrotron-producing jets require fundamentally different shock conditions than purely thermal ones. The high sensitivity and resolution needed for the science goals make JWST the only facility capable of these observations.

OBSERVING DESCRIPTION

We propose to observe the massive star-forming region RAFGL2591, using NIRCam and NIRSpec-IFU. This will cover the jet lobes, collimated jet and shocked regions of the target source. The main objective of these observations will be to provide high sensitivity spectra of the jet lobes and high resolution continuum subtracted images of the protostellar jet and shocked regions.

We will use NIRCam with the F212N and F164N narrow-band filters, together with the medium-band F210M and F162M filters. For NIRSpec-IFU, we will employ the G140H/F100LP and G235H/F170LP disperser–filter combinations, providing wavelength coverage from 0.97 to 3.17 μm at a spectral resolution of R 2700.

The integration time was set in order to detect lines of about ~ 3×10^{1} erg s^{1} cm^{2} $pixel^{1}$ with a S/N > 10, according to the JWST ETC. Such a limit has been defined on the basis of the expected brightness of key diagnostic lines located at different wavelengths across the observed spectral range, such as [SII] and [NI] lines, predicted with our simulated spectrum of the jet emission.

With NIRCam, the requested sensitivity is obtained with 4 integrations of 8 groups each. The RAPID readout pattern is selected to provide enough groups per integration for an optimised pipeline calibration. We use noiseless sky background for an optimal background subtraction.

For NIRSpec, the requested sensitivity limit will be obtained with 3 integrations of 8 groups in a 4-POINT-DITHER pattern. We will use the NRS readout mode.

Proposal 10108 - Targets - Probing shock driven particle acceleration using near infrared observations

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	East_Lobe	RA: 20 29 28.5462 (307.3689425d) Dec: +40 11 13.59 (40.18711d) Equinox: J2000	Epoch of Position: 2000	
Comments: *Category=ISM* *Description=[Herbig-Haro objects, Protostars, Stellar jets]* *Extended=YES*				
(2)	West_Lobe	RA: 20 29 21.9690 (307.3415375d) Dec: +40 11 11.94 (40.18665d) Equinox: J2000	Epoch of Position: 2000	
Comments: *Category=ISM* *Description=[Herbig-Haro objects, Protostars, Stellar jets]* *Extended=YES*				
(3)	East-West_lobes	RA: 20 29 25.4199 (307.3559163d) Dec: +40 10 43.30 (40.17869d) Equinox: J2000	Epoch of Position: 2000	
Comments: *Category=ISM* *Description=[Herbig-Haro objects, Protostars, Stellar jets]* *Extended=YES*				

Observation

Proposal 10108, Observation 1: NIRcam H and [Fe II]

Fri May 22 15:01:53 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Comments: The position angle ranges are selected such that the bright central source falls within the chip gap, while the shocked, extended emission of interest is positioned on the active detector area, avoiding the chip gaps.

Diagnostics

(Visit 1:1) Warning (Form): Data Excess over lower threshold

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	East-West_lobes	RA: 20 29 25.4199 (307.3559163d) Dec: +40 10 43.30 (40.17869d) Equinox: J2000	Epoch of Position: 2000	

Comments:
Category=ISM
Description=[Herbig-Haro objects, Protostars, Stellar jets]
Extended=YES

Template

Module	Subarray
B	FULL

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	NONE		STANDARD		3

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F212N	F300M	RAPID	8	4	12	3	1127.361	
2	F164N+F150W2	F250M	RAPID	8	4	12	3	1127.361	
3	F210M	F356W	RAPID	8	2	6	3	547.575	
4	F162M+F150W2	F277W	RAPID	8	2	6	3	547.575	

Special Requirements

Aperture PA Range 87.05262691 to 93.05262691 Degrees (V3 87.0 to 93.0)
Aperture PA Range 177.05262691 to 183.05262691 Degrees (V3 177.0 to 183.0)
Aperture PA Range 267.05262691 to 273.05262691 Degrees (V3 267.0 to 273.0)
Aperture PA Range 357.05262691 to 3.05262691 Degrees (V3 357.0 to 3.0)

Observation

Proposal 10108, Observation 2: NIRSpec IFU for east lobe

Fri May 22 15:01:53 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Comments: The position angle constraints are selected to ensure that the bright central source lies outside the MOS field, thereby reducing contamination from leakage through open shutters.

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	East_Lobe	RA: 20 29 28.5462 (307.3689425d) Dec: +40 11 13.59 (40.18711d) Equinox: J2000	Epoch of Position: 2000	

Comments:
Category=ISM
Description=[Herbig-Haro objects, Protostars, Stellar jets]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140H/F100LP	NRS	8	3	false	true	NONE	4	12	4251.761	
2	G235H/F170LP	NRS	8	3	false	true	NONE	4	12	4251.761	

Special Requirements

Aperture PA Range 109.97164917 to 259.97164917 Degrees (V3 331.0 to 121.0)
Aperture PA Range 356.97164917 to 8.97164917 Degrees (V3 218.0 to 230.0)

Observation

Proposal 10108, Observation 3: NIRSpec IFU for west lobe

Fri May 22 15:01:53 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Comments: The position angle constraints are selected to ensure that the bright central source lies outside the MOS field, thereby reducing contamination from leakage through open shutters.

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	West_Lobe	RA: 20 29 21.9690 (307.3415375d) Dec: +40 11 11.94 (40.18665d) Equinox: J2000	Epoch of Position: 2000	

Comments:
Category=ISM
Description=[Herbig-Haro objects, Protostars, Stellar jets]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140H/F100LP	NRS	8	3	false	true	NONE	4	12	4251.761	
2	G235H/F170LP	NRS	8	3	false	true	NONE	4	12	4251.761	

Special Requirements

Aperture PA Range 148.97164917 to 183.97164917 Degrees (V3 10.0 to 45.0)
Aperture PA Range 293.97164917 to 38.97164917 Degrees (V3 155.0 to 260.0)